

Hybrid Neuro-Symbolic Deliberative Solver (HNS-DS): A Verified Architecture for Human-Like Problem Solving in Software and Mathematics

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Abstract

Large language models produce fluent code and mathematical text but lack structured deliberation and formal guarantees. We present the *Hybrid Neuro-Symbolic Deliberative Solver (HNS-DS)*, a formal architecture combining hierarchical planning, program synthesis (CEGIS-style), symbolic solvers (SMT/CAS/CP-SAT), and a verification-driven learning loop. We define tasks as $T = (\Sigma, S, V)$ (specification, candidate space, verifier), present the RIR-RI algorithm (Reinforced Iterative Repair with Retrieval & Induction), and provide rigorous proofs of soundness and bounded completeness. We further model the iterative refinement as an absorbing Markov chain and derive spectral bounds for expected convergence under a stochastic selection policy. TikZ diagrams illustrate the pipeline and loop; an appendix contains a reproducible implementation checklist and starter-repo blueprint. The paper provides a mathematical and engineering foundation for building verifiable, human-like problem solvers.

1 Introduction

Human problem solving in mathematics and software engineering is iterative, hypothesis-driven, and verification-centered. Modern generative models (LLMs) approximate aspects of this behavior by producing plausible text and code, but they cannot guarantee correctness or produce verifiable proofs

in general. We propose an alternative: a hybrid, deliberative architecture that synthesizes solutions symbolically, verifies them, and learns heuristics from episodes.

This paper:

- Formalizes the HNS-DS architecture and the problem model.
- Presents the RIR-RI algorithm with pseudocode and invariants.
- Proves soundness and bounded completeness.
- Models the refinement as an absorbing Markov chain and gives expected-iteration bounds under reasonable stochastic assumptions.
- Provides diagrams, a reproducible evaluation plan, and an implementation appendix.

2 Related Work

We summarize the most relevant literature: CEGIS and program synthesis [4], SMT and automated reasoning (Z3) [1], CAS systems (SymPy), neuro-symbolic research [5], and automated program repair (GenProg) [3].

3 Formal model

We define the computational model used by HNS-DS.

Definition 3.1 (Task). A *task* T is a triple $T = (\Sigma, S, V)$ where:

- Σ is a formal specification (logical constraints, pre/postconditions, and/or input-output examples).
- S is a (possibly implicit) search space of candidate solutions (programs, proofs).
- $V : S \rightarrow \{0, 1\}$ is a verifier: $V(s) = 1$ iff $s \models \Sigma$ (i.e., s satisfies Σ).

Remark 3.2. In practice, Σ can combine executable tests (unit/integration tests), SMT-expressible invariants, and algebraic targets. S is represented by grammars, sketches, or templates.

3.1 Components

HNS-DS uses the following modules (notation used later):

$$\{P, F, W, PL, SO, SY, V, L\}$$

Perception P parses input; Formalizer F constructs Σ ; World model W stores ASTs and artifacts; Planner PL decomposes goals into subgoals; Solver Orchestrator SO selects solvers; Synthesizer SY proposes candidates; Verifier V checks correctness; Learner L updates heuristics.

4 Algorithm: RIR-RI (pseudocode and invariants)

We present the central algorithm, which runs per task.

4.1 RIR-RI pseudocode

Input: Task $T=(,S,V)$, budget B (per subgoal)

Output: Verified solution s^* or FAIL

```
1.  $\leftarrow F(P(\text{raw\_input}))$ 
2.  $G \leftarrow PL()$  // decompose into subgoals  $g1..gm$ 
3. For each  $gi$  in  $G$ :
4.    $Ei \leftarrow \text{initial\_examples}(gi)$  // may be empty
5.   while budget not exhausted:
6.      $s\_hat \leftarrow SY(gi, Ei, W, L)$  // propose candidate
7.      $\text{pass}, ce \leftarrow V.\text{verify}(s\_hat, )$  // run tests/SMT/CAS
8.     if pass:
9.       accept  $s\_hat$  for  $gi$ ; break
10.    else
11.       $Ei \leftarrow Ei \cup \{ce\}$  // add counterexample
12.  if no candidate found  $\rightarrow$  return FAIL
13. Compose accepted sub-solutions into  $S\_global$ 
14.  $\text{final\_pass}, ce\_final \leftarrow V.\text{verify}(S\_global, )$ 
15. if  $\text{final\_pass}$  return  $S\_global$  else return FAIL
```

4.2 Invariants and notes

- Each added counterexample reduces the feasible candidate set for that subgoal (monotonic refinement).

- The verifier V may be a suite: unit tests, SMT checks, CAS simplifications.
- The learner L may alter heuristic order (prioritizing likely candidates).

5 Theoretical guarantees

We now prove key properties under explicit assumptions.

5.1 Soundness

Theorem 5.1 (Soundness). *If RIR-RI returns solution s^* , then $V(s^*) = 1$.*

Proof. By algorithm line 14–15, the algorithm returns only after V verifies the composed solution. Hence the returned s^* satisfies Σ by definition of V . $\square$

5.2 Bounded completeness

Theorem 5.2 (Bounded completeness). *Assume each subgoal g has finite candidate search space S_g . If there exists a global solution s^* with $V(s^*) = 1$, and the synthesizer enumerates or can generate all candidates in S_g given unbounded budget, then RIR-RI with unbounded budget will find s^* .*

Proof. Under the finiteness assumption, each counterexample eliminates a non-empty subset of the current feasible set. The feasible set cannot be reduced indefinitely without reaching a fixed point. Because the true solution s^* remains feasible (it satisfies all counterexamples), eventually enumeration reaches s^* (or an equivalent verified candidate) and the verifier will accept. This is the standard termination argument for CEGIS over finite domains (see [4]). $\square$

5.3 Convergence as fixed point

The refinement operator on feasible sets is monotone and decreasing; repeated application reaches a fixed point corresponding to the verified solutions. Thus the algorithm converges in finitely many steps for finite candidate spaces.

6 Stochastic selection and spectral analysis

Synthesis often involves heuristics or randomized candidate selection. We model the solver’s candidate-selection as a stochastic process and derive expected convergence bounds.

6.1 Model

Enumerate a (finite) set of candidates $S = \{s_1, \dots, s_n\}$. Let $\mathbf{x}_t \in \{0, 1\}^n$ indicate active candidates at iteration t . Instead of a deterministic transition, suppose candidates are sampled according to a probability distribution π_t over active candidates. Let $p_{i \rightarrow j}(t)$ denote the probability that when candidate s_i is tried and fails, the resulting refinement leaves s_j still active. We aggregate these into a (time-dependent) transition matrix $P_t \in [0, 1]^{n \times n}$.

6.2 Absorbing Markov chain

If verified solutions correspond to absorbing states (once verified they remain), then P_t approximates a Markov chain with absorbing set $A \subseteq S$. Standard Markov chain theory gives expected absorption times in terms of the fundamental matrix $N = (I - Q)^{-1}$, where Q is the transient-states submatrix [2]. The expected number of candidate trials until absorption starting from initial distribution π_0 is bounded by entries of N .

6.3 Spectral gap heuristic

Under simplifying assumptions of time-homogeneity and irreducibility among transient states, the spectral gap $\gamma = 1 - \lambda_2$ (where λ_2 is second-largest eigenvalue magnitude) controls mixing time. Larger γ implies faster mixing and typically faster expected absorption. Practically, the learner L aims to increase effective γ by prioritizing candidates with higher elimination power.

7 Matrix diagnostics and visualization

We recommend the following diagnostics:

- Visualize transition matrices P_t or pruning matrices as sparse heatmaps.
- Track candidate-embedding similarity matrices S (candidates vs counterexamples).
- Plot absorption-time histograms across tasks and seed randomizations.

8 Diagrams

Below are TikZ diagrams included in the LaTeX source. They render the pipeline, CEGIS loop, and absorbing chain.

9 Evaluation Plan (reproducible)

We outline an experiment plan you can run to validate HNS-DS.

9.1 Benchmarks

- **Defects4J** for Java program repair.
- **HumanEval** for Python function synthesis.
- **MiniF2F** for formal math proofs.

9.2 Metrics

- Success rate (verified solutions / tasks).
- Time to verification (wall-clock).
- Iteration count (synth-verify cycles).
- PR acceptance rate (for code patches after human review).
- Learning curve (success vs number of episodes).

9.3 Ablations

Run baseline comparisons:

1. HNS-DS with Learner turned off.
2. HNS-DS with only enumerative synthesis.
3. HNS-DS with no planner (monolithic synthesis).

10 Implementation appendix: starter blueprint

This appendix gives a practical checklist and a minimal repository layout to implement HNS-DS.

10.1 Checklist

1. **Perception / Formalizer:** implement a rule-based $NL \rightarrow \Sigma$ front-end; start with template-based specs.
2. **World model:** implement AST extraction, call graphs, and test mapping.
3. **Synthesizer:** implement small enumerative engine + CEGIS loop. Use Python for prototyping.
4. **Verifier:** integrate pytest for runtime checks and Z3 for SMT checks.
5. **Learner:** collect episodes (JSONL) and implement rudimentary reranker (sentence-transformers + FAISS).
6. **Orchestration:** implement the RIR-RI loop in a single Python orchestrator.

7. **Sandboxing:** use Docker to run tests safely.
8. **Logging / Reproducibility:** store episodes and random seeds.

10.2 Repository blueprint

```
hnsds/  
  orchestrator.py  
  perception/  
    parser.py  
  formalizer/  
    spec_builder.py  
  planner/  
    htn_planner.py  
  synthesizer/  
    enumerative.py  
    sketch_interface.py  
  verifier/  
    pytest_runner.py  
    z3_interface.py  
  learner/  
    episode_logger.py  
    reranker_train.py  
  sandbox/  
    docker_runner.py  
  experiments/  
    run_defects4j.sh  
    run_humaneval.sh
```

11 Discussion and limitations

HNS-DS offers formal guarantees when specifications are formalizable and search spaces are finite. Practical limitations include:

- The $NL \rightarrow \Sigma$ mapping: ambiguous natural language requires human-in-loop or trained parsers.
- Search complexity: worst-case exponential; mitigated by heuristics and solver constraints.
- Integration engineering: combining many solvers and tools requires careful engineering.

12 Conclusion

We presented HNS-DS, a rigorously defined hybrid architecture for deliberative, verifiable problem solving. We provided algorithms, proofs, spectral analysis, diagrams, and an evaluation plan. This is a practical and mathematical foundation for reliable AI developer assistants.

Acknowledgments

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References

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